

frontmatter — build plan

Work breakdown, critical path, and eight milestones whose definitions of done are all executable.

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§0 How to use this

This plan is executed through `specs/`, not through this document. A lane is not "done" because it feels done; it is done when its spec reaches `state: verified`, which the harness will only write after every `verify:` command exits 0 and a red proof exists that fails against the unfixed code.

```
npm run spec      # the contract gate
npm run corpus    # the fidelity corpus, 8,513 files, byte-pinned
npm run verify    # typecheck → lint → test → build → arch → spec
```

Three rules that govern every unit below.

1. **Red proof before green.** A test on a rare fault proves nothing until it fails against the unfixed code. If you cannot make it fail, the test does not cover the bug — record that in the spec's `## Open` rather than reporting a pass.
 2. **Agents propose, the founder merges.** Capture `git rev-parse HEAD` before and after every agent run and reconcile the delta. The instruction not to commit is advisory; the reconciliation is the gate.
 3. **Re-derive every number at write time.** The PRD's §32 exists because we did not.
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§1 Decisions that unblock everything

These were open. They are now answered with reasoning, and each has an "evidence that would flip it" so the answer stays falsifiable.

#	DECISION	ANSWER	WHY IT HOLDS
A1	Sync engine	Git-as-sync, server-authoritative at the commit boundary. ~80% already built: <code>merge3</code> , optimistic concurrency, <code>/api/vault/merge</code>	A CRDT must never own the file's bytes. "Every untouched byte bit-identical" is unassertable across a CRDT merge, because the result derives from the operation graph, not the bytes. Adopting a CRDT adopts precisely the architectural gap we cite against OpenKnowledge. If per-keystroke co-editing is ever required, run it as an ephemeral session whose only output is one splice at session end — a scratch buffer, never a store of record
A1b	TO's exit condition	Rewritten. From "two-device convergence" to " zero bytes lost, every divergence surfaced as a reviewable hunk "	"Convergence" is a CRDT word; the architecture that must satisfy it is <code>merge3</code> + OCC. The PRD promised a property its own chosen architecture does not provide. This is a new contradiction, not the one D2 named
A2	Where state lives	Git repo = document truth, never moves. One Postgres as control plane only, holding zero document bytes. R2 for attachments >1 MB and derived artifacts	Publish-slug uniqueness currently works by scanning the whole snapshot. Identity, entitlements, billing, audit have no home at all
A3	Multi-tenancy	<code>workspace_id</code> on every row, pooled RLS. GitHub App installation is the connector , never the tenant identity	Users connect two repos, transfer repos, and uninstall. Identity must survive all three. Retrofitting <code>workspace_id</code> after launch is the highest-cost change on the board
A5	Search	Stop shipping the whole snapshot → Postgres FTS + trigram as the server tier → MiniSearch client-side only. CJK via a bigram tokenizer , not a dictionary segmenter	77.5 MB parsed per cold start, and the snapshot response is 16.4× the platform body cap. No stateful search cluster before 1,000 tenants
A14	MDMAX wiring	Four seams, in order: ingress gate (live) → write gate → splice engine → certificate. MDMAX exports pure functions over bytes and imports nothing from <code>src/modules</code>	Currently 1 symbol from 1 of 13 files is used. Do not wire the write gate before NF-1/NF-3 — at the measured refusal rate it would reject 83% of foreign publishes, an availability incident wearing a correctness costume
R-carrier	How a render profile is written on disk	Callout <code>> [!kind]</code> for prose, fenced code for opaque data. Nothing else. <code>:::</code> accepted on input, normalised away on save	Measured across four engines plus GitHub live: an unclosed fence swallows the rest of the document (CommonMark §4.5, spec-mandated); a callout has no closing marker to lose . This corrects PRD §9
A15.1	Dual identity	Collapse to one. Keep NextAuth + the GitHub App; delete Firebase	Two auth paths is two session-fixation surfaces for one operator, plus an unused Firestore import in the client bundle

Still yours to decide, and each re-scopes a lane: D3 the name (see §7), D8 do documents leave the device, D9 BYO vs platform key, D14 does the agent get write authority, D15 publishing in v1 or v2, D17 CJK in or out.

§2 Work breakdown

Points: XS=1 · S=2 · M=4 · L=8 · XL=16 . One XL ≈ one substantial shipped surface ≈ 14–21 calendar days solo, so 1 pt ≈ 1.09 calendar days. Calendar days, not working days — the measured 25% active-day density on this branch is *inside* that rate, not a multiplier on it.

R0 — engine truth · 58 pts · the declared first lane

ID	DELIVERABLE	SIZE	SPEC
R0.1	NF-1 — a <code>-</code> item at column 0 is a continuation of the preceding key	M	<code>engine/nf-001-zero-indent-sequence</code>
R0.2	NF-2 — flow-seq closing <code>]</code> at column 0	S	—
R0.3	NF-3 — <code>FM_OPEN</code> misses a bare <code>---\r</code> ; a lone set prepends a second frontmatter block. Set-destructive and invisible to a round-trip oracle. Red proof first	M	to write
R0.4	NF-4 — quoted-key <code>SAFE_KEY</code> . Blocked on a decision , not on code: what key equality means for NFC vs NFD	L	founder
R0.5	CI. Port md's <code>ci.yml</code> ; all four harness scripts already exist	S	<code>platform/ci</code>
R0.6	Foreign-corpus standing gate	—	DONE — 8,513 files pinned, verify red-proofed
R0.7	Six construct-detector defects	M	—
R0.8	CJK — <code>countWords</code> segmenter, MiniSearch bigram tokenizer	M	gated on D17
R0.9	Wire MDMAX seams 1–3	L	<code>engine/mdmax-seams</code>
R0.10	Reconcile <code>globals.css</code> with the design system	M	—
R0.11	mdmax audit Tier 3/4 residue	M	—
R0.12	Replace the <code>npm run budget</code> stub with a real byte ceiling	S	—
R0.13	Close the open decisions. D5 PAT rotation is today, not a milestone	M	founder

The rest

LANE	PTS	CONTENTS
T0 Trust surface	32	sync chip <code>S</code> · conflict inbox <code>M</code> · named-version history <code>M</code> · since-you-last-opened banner <code>S</code> · background auto-sync <code>L</code> · local history + section restore <code>L</code> · the two-device rig <code>M</code>
T1 Tenancy + launch	42	identity + multi-tenancy <code>XL</code> · GitHub App <code>L</code> · multi-vault <code>M</code> · mobile pass <code>M</code> · HOME.md <code>M</code> · quick capture <code>S</code> · companion plugin <code>M</code>
T3 AI protocol	44	MCP server + <code>land()</code> <code>L</code> · review loop <code>L</code> · provenance + implicit telemetry <code>M</code> · differ <code>M</code> · cert distribution <code>L</code> · citation-gated answers <code>M</code> · session continuity <code>M</code> · publish the session format <code>M</code>
T4 Capture	20	chat-side skill + paste inbox <code>M</code> · ChatGPT/Claude importers <code>L</code> · promotion loop <code>M</code> · retro-capture <code>M</code>
T2 Renders	42	out of v1 except two <code>XS</code> cleanups: the <code>/language-(\w+)/</code> hyphen fix and deleting dead <code>editable-table.tsx</code>
INFRA/LEGAL	25	MoR before the first <i>paid</i> signup · EU Art. 27 rep before the first EU <i>free</i> signup · CA + lawyer · ToS/privacy · auth, error tracking, R2 — all buy
GTM	34	land Markex's tree (RULE-2 gated) · LinkedIn Documents-API probe <code>S</code> · post-as-document <code>L</code> · launch calendar in-product <code>M</code> · 20 posts <code>XL</code>

§3 Critical path

```
R0.3 → R0.1 → R0.2 → R0.4 → R0.9 → T0(auto-sync → local history → two-device rig)
    → T1(identity → GitHub App → HOME.md) → T3(MCP+land()) → review loop → cert distribution)
    → T4(importers → promotion loop) → v1
```

- **R0.5 (CI) is not on the path but gates the credibility of everything after it.** Do it in the first 48 hours so every later unit lands under a gate.
 - **T2 is off the path entirely.** Its only path-touching items are the two `XS` cleanups.
 - **Legal items are date-gates, not effort-gates.** They are other people's calendars.
 - **R0.4 is the one engine unit an agent cannot start** — it needs a written decision on Unicode key equality first.
-

§4 Milestones — every definition of done is executable

M	GATE	DONE MEANS
M0	CI live	A PR runs typecheck·lint·test·build·arch·spec and fails on a deliberately broken commit . <code>npm run budget</code> asserts a real byte ceiling, not <code>echo</code> . One red run is required before the first green is trusted
M1	Engine truth	Over the pinned corpus: refused ≤ 2 of 7,959 (from 6,614), changed = 0, threw = 0 . NF-3's set-only assertion fails on <code>HEAD~1</code> and passes on <code>HEAD</code> . <code>date created set + rename</code> succeed on ≥ 812 files. <code>npm run corpus</code> exits 0 at gate time
M2	MDMAX wired	Seams 1–3 live; ≥ 8 of the 12 unreachable mdmax files have a product importer; <code>npx mdmax cert <file> --fail-on=BROKEN</code> exits 1 on a seeded BROKEN fixture inside CI
M3	Trust surface	Two devices, both offline, both edit the same file, both reconnect: zero bytes lost and every divergence surfaced as a reviewable hunk , watched by a human, recorded. Section-level restore returns the file to a prior sha with every untouched byte identical
M4	Tenancy	A second GitHub account, never used in development, signs in via the GitHub App , connects one repo, edits, commits — no founder intervention, no shared secret
M5	Agent protocol	An external agent edits a real vault through <code>land()</code> ; every change appears in the review surface and is rejectable; a rejected change leaves the file byte-identical to pre-edit
M6	Capture	One ChatGPT export ZIP → durable typed documents in one click, and the verification report enumerates every dropped construct by count
M7	v1 public	M0–M6 green + MoR live + EU rep appointed + ToS published + one paying non-founder account

Reject these if proposed as a DoD: "kanban feels good" · "import works" · "fidelity is high" · anything whose evidence is a screenshot · anything a `grep` over source can satisfy.

§5 Who does what

AN AI AGENT CAN OWN (DISJOINT ARTIFACTS)	ONLY THE FOUNDER CAN DO
NF-1/NF-2 walk change + fixtures	The NF-4 key-equality decision
Corpus runner and gates	All open decisions
The six construct detectors, one agent each	PAT rotation
CJK segmenter + tokenizer	GitHub App registration, secrets, callback
CSS reconciliation	MoR, CA, lawyer, EU representative
Tier 3/4 residue	Watching the M3 two-device run
Test authoring for every unit	Accepting or rejecting every agent diff
Doc and spec sync	Any RULE-2 operation

Do not parallelise: NF-1 and NF-4 against the same file — both touch the key walk. And do not fan out T0's sync work; it is one creative target carrying implicit architecture decisions.

§6 Calendar

Assumption stated so it can be disagreed with: 1 pt = 1.09 calendar days, from 2026-08-29.

MILESTONE	CUM. PTS	DATE
R0 complete	58	2026-10-31
+ T0	90	2026-12-05
+ T1	132	2027-01-20
+ T3 (v1 subset)	176	2027-03-09
+ T4 = code-complete	196	2027-03-31
+ legal/infra = shippable	221	2027-04-28
+ full GTM	255	2027-06-04

R0 alone is **9 weeks**. That is consistent with the PRD's own "multi-month R0 lane before any customer-visible feature", and it is the part most likely to be wished away.

§7 The ten things most likely to blow this

#	RISK	EARLY WARNING – CHECK WEEKLY
1	"NF-1 recovers 99.98%" is an inference from bucketing refusal causes, not a measured result of the patched writer	First patched run refuses >10 files. If the residual is >2, NF-1 was never one bug
2	Corpus loss or drift	Any run whose file count ≠ 8,513. Mitigated: pinned and red-proofed
3	NF-4 is a design task wearing a regex costume	Two weeks pass with no written NFC/NFD decision
4	"Wire MDMAX in" is an integration surface, not a wiring task — 12 files with zero call sites	Week 2 of R0.9 still has 0 new product importers. Count them
5	CI's first green will be false. <code>budget</code> is <code>echo</code> , <code>e2e</code> is <code>if: false</code> , and four gates in this repo already reported green while blind	CI passes on a deliberately broken commit
6	Cadence is bursty — 8 active days in 32	Two consecutive weeks with zero commits
7	T1 identity is the only XL, scored from analogy not from this repo	A week of schema churn instead of a signed-in second account
8	Open decisions gate scheduled work; D8 alone can delete T3	Any lane started before its gating decision is written down
9	Legal lead times are vendor-clock	EU-reachable free tier with no Art. 27 representative
10	The name is unresolved	Any brand spend before D3 closes. See below

On the name. The strongest candidate found is `stetfile` — *stet* is the one word in English meaning "let it stand, preserve the original exactly", which is literally the engine's contract. Clean on all four measured axes: npm 404, GitHub 404, zero repos, no product. Tracked risk: an agent-oriented package `stetmark` shipped 2026-08-05 with ~5,753 downloads/month, so bare `stet` branding is contested even though `stetfile` is not.

If you keep "frontmatter", the honest finding is that the collision is **narrower than the raw install count suggests** — Front Matter CMS is a VS Code *extension*, a plugin inside another editor, at 2,539 stars. But the compromise to avoid is explicit: **do not ship "Frontmatter" bare while quietly holding qualified handles.** That takes the full trademark and SEO exposure and buys nothing. Either qualify the spoken name or leave the word. Nothing here is trademark clearance; that needs counsel.

§8 The first week, concretely

DAY	DO	PROVES
1	Rotate the two PATs. Then CI: port <code>ci.yml</code> , add <code>spec</code> and <code>corpus</code> to the workflow	Only you can do the first. The second gates everything after it
1	Make CI fail on purpose, once, before trusting any green	That the gate can see
2	NF-3 red proof — a set-only assertion that fails on today's code	Set-then-delete cancels out; that is how NF-3 stayed invisible
2-4	NF-3 fix, then NF-1 red proof and fix	The 83% refusal rate is the number the whole fidelity claim rests on
4	Re-derive the recovery rate from the patched writer. Replace the 99.98% inference with a measurement	Risk #1
5	Write the NF-4 decision — one page on NFC/NFD key equality	Unblocks the only agent-blocked unit
5	The two <code>XS</code> cleanups: the <code>/language-(\w+)/</code> hyphen fix, delete <code>editable-table.tsx</code>	Unblocks every render profile later
Ongoing	One spec per unit, <code>draft → verified</code> , never hand-written	The plan is executed through <code>specs/</code> , or it is not executed

The engine work is unglamorous and it is first for a reason: every fidelity number we intend to market is currently false at the measured refusal rate, and publishing before the fix converts a bug into a public claim.